

Supplementary Material

1 Supplementary Text 1. Search strategies

Pubmed n=354

1, "Dyslipidemias"[MeSH Terms] OR "Dyslipidemia"[Title/Abstract] OR "Dyslipoproteinemias"[Title/Abstract] OR "Dyslipoproteinemia"[Title/Abstract] OR "Atherosclerosis"[MeSH Terms] OR "Atheroscleroses"[Title/Abstract] OR "Atherogenesis"[Title/Abstract]

2,"alirocumab"[Supplementary Concept] OR "SAR236553"[Title/Abstract] OR "SAR-236553"[Title/Abstract] OR "REGN-727"[Title/Abstract] OR (("antibodies, monoclonal"[MeSH Terms] OR ("antibodies"[All Fields] AND "monoclonal"[All Fields]) OR "monoclonal antibodies"[All Fields] OR ("monoclonal"[All Fields] AND "antibody"[All Fields]) OR "monoclonal antibody"[All Fields]) AND "REGN727"[Title/Abstract]) OR "reg727 monoclonal antibody"[Title/Abstract] OR "REGN727"[Title/Abstract] OR "praluent"[Title/Abstract] OR "evolocumab"[Supplementary Concept] OR "Repatha"[Title/Abstract] OR "AMG-145"[Title/Abstract] OR "AMG-145"[Title/Abstract]

3, "randomized controlled trial"[Publication Type] OR "controlled clinical trial"[Title/Abstract] OR "random allocation"[Title/Abstract] OR "double-blind"[Title/Abstract] OR "single blind"[Title/Abstract] OR "Placebo"[Title/Abstract] OR "Randomly"[Title/Abstract] OR "Randomized"[Title/Abstract] OR "clinical trial*"[Title/Abstract] OR "trial*"[Title/Abstract] OR "RCT"[Title/Abstract] OR "random*"[Title/Abstract]

4, #1 AND #2 AND #3

WOS n=286

1, TS= (Dyslipidaemia OR Dyslipidemia OR Dyslipoproteinemias OR Dyslipoproteinemia OR Atherosclerosis OR Atheroscleroses OR Atherogenesis)

2, TS=(alirocumab OR SAR236553 OR SAR-236553 OR REGN-727 OR monoclonal antibody REGN727 OR REGN727 monoclonal antibody OR REGN727 OR praluent OR evolocumab OR Repatha OR AMG-145 OR AMG 145)

3, TS=(Randomized Controlled Trial OR controlled clinical trial OR random allocation OR double-blind OR single- blind OR Placebo OR Randomly OR randomized OR clinical trial* OR trial* OR RCT OR Random*)

4, #1 AND #2 AND #3

Cochrane Library n=223

1, (Dyslipidaemia OR Dyslipidemia OR Dyslipoproteinemias OR Dyslipoproteinemia OR Atherosclerosis OR Atheroscleroses OR Atherogenesis) :ti,ab,kw

2, (alirocumab OR SAR236553 OR SAR-236553 OR REGN-727 OR monoclonal antibody REGN727 OR REGN727 monoclonal antibody OR REGN727 OR praluent OR evolocumab OR Repatha OR AMG-145 OR AMG 145):ti,ab,kw

3, (Randomized Controlled Trial OR controlled clinical trial OR random allocation OR double-blind OR Placebo OR Randomly OR randomized OR clinical trial* OR trial* OR RCT OR Random*):ti,ab,kw

4, #1 AND #2 AND #3

Embase n=875

1, 'dyslipidaemia'/exp OR 'dyslipidaemia' OR 'Dyslipidemia':ab,ti OR 'Dyslipoproteinemias':ab,ti OR 'Dyslipoproteinemia':ab,ti OR 'Atherosclerosis'/exp OR 'Atherosclerosis' OR 'Atheroscleroses':ab,ti OR 'Atherogenesis':ab,ti

2, 'alirocumab'/exp OR 'alirocumab' OR 'sar236553':ab,ti OR 'sar-236553':ab,ti OR 'regn-727':ab,ti OR 'monoclonal antibody regn727':ab,ti OR 'regn727 monoclonal antibody':ab,ti OR 'regn727':ab,ti OR 'praluent':ab,ti OR 'evolocumab'/exp OR evolocumab OR 'repatha':ab,ti OR 'amg-145':ab,ti OR 'amg 145':ab,ti

3, 'Randomized Controlled Trial'/exp OR 'Randomized Controlled Trial' OR 'controlled clinical trial':ab,ti OR 'random allocation':ab,ti OR 'double-blind':ab,ti OR 'single-blind':ab,ti OR 'Placebo':ab,ti OR 'Randomly':ab,ti OR 'randomized':ab,ti OR 'clinical trial*':ab,ti OR 'trial*':ab,ti OR 'RCT':ab,ti OR 'Random*':ab,ti

4, #1 AND #2 AND #3

2 Supplementary Table 1. The pooled CERs of PCSK9 inhibitors.

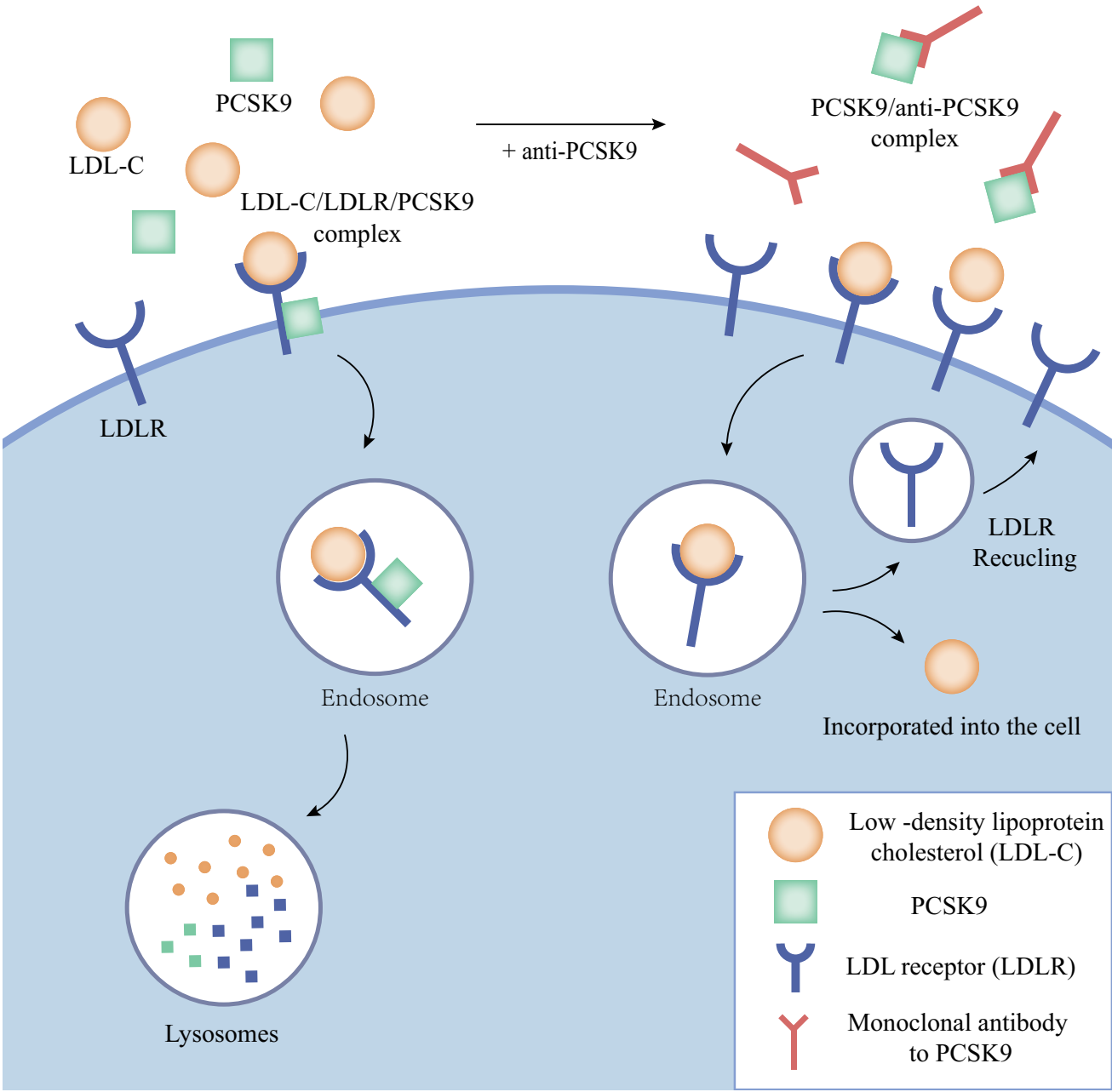
Outcomes	Drug type	CER (control event rate)
major vascular events	PCSK9 inhibitors	0.165
	alirocumab	0.190
	evolocumab	0.147
all-cause mortality	PCSK9 inhibitors	0.032
	alirocumab	0.037
	evolocumab	0.029
cardiovascular death	PCSK9 inhibitors	0.021
	alirocumab	0.026
	evolocumab	0.017
myocardial infarction	PCSK9 inhibitors	0.054
	alirocumab	0.069
	evolocumab	0.044
stroke	PCSK9 inhibitors	0.017
	alirocumab	0.014
	evolocumab	0.018
coronary revascularization	PCSK9 inhibitors	0.075
	alirocumab	0.081
	evolocumab	0.071

3 Supplementary Table 2. Univariate meta-regression analysis of PCSK9 inhibitors on major vascular event

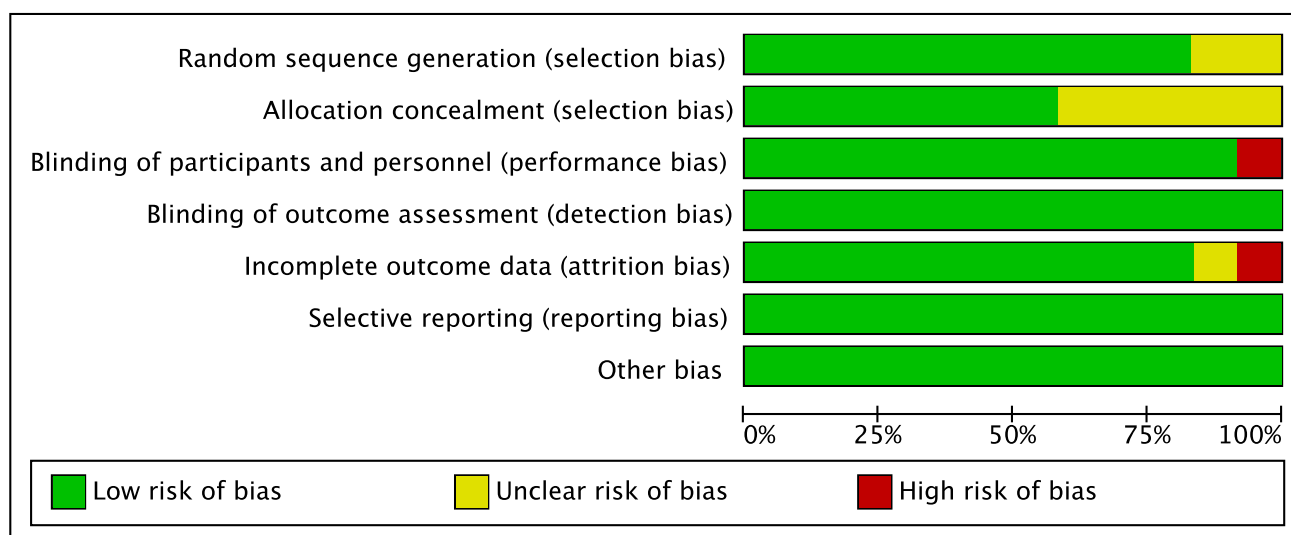
Model	Covariate	Classification	No. of Studies	<i>P</i> Value for Heterogeneity
1	Type of PCSK9 inhibitors	Alirocumab	8	0.221
		Evolocumab	4	
2	Mean age, y	< 60	8	0.419
		≥60	4	
3	Published year		14	0.274
4	Men, %		14	0.219
5	Follow-up, y		14	0.380
6	DM, %		14	0.333
7	CAD, %		14	0.293
8	statins, %		14	0.546

DM, Diabetes mellitus; CAD, coronary artery disease

4 **Supplementary Figure 1.**



Supplementary Figure 1. The mechanisms of action of PCSK9 inhibitors.

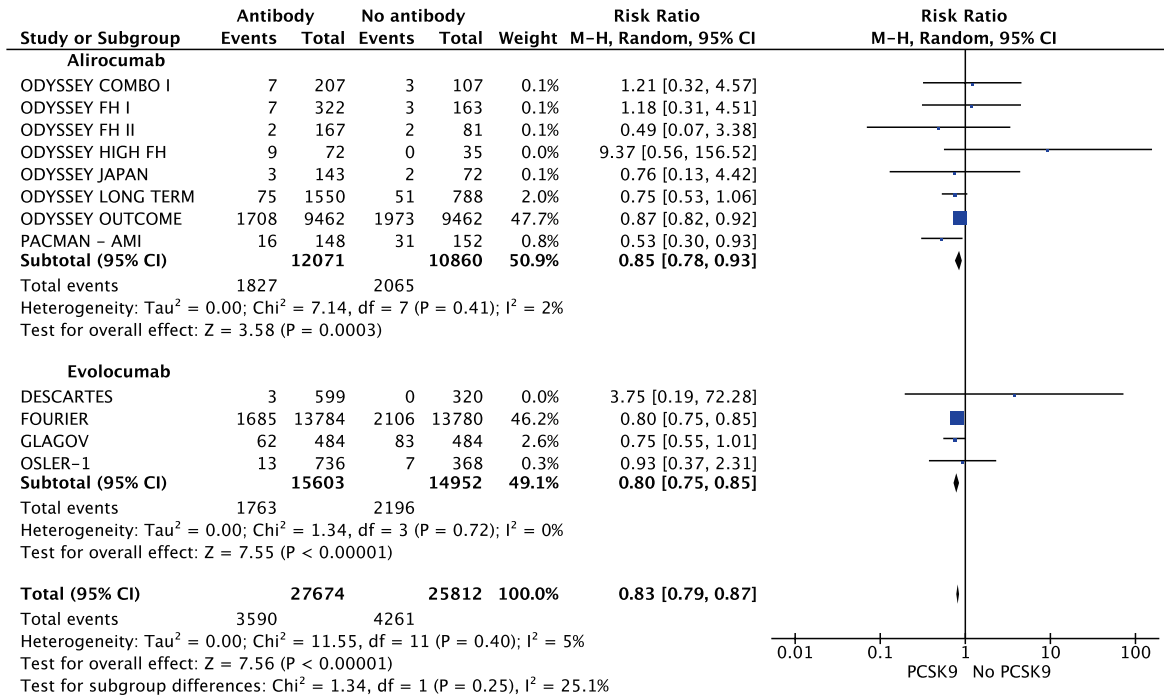
5 **Supplementary Figure 2.**

	PACMAN - AMI	OSLER-1	ODYSSEY OUTCOME	ODYSSEY LONG TERM	ODYSSEY JAPAN	ODYSSEY HIGH FH	ODYSSEY FH II	ODYSSEY FH I	ODYSSEY COMBO I	GLAGOV	FOURIER	DESCARTES	
Random sequence generation (selection bias)	+	+	?	+	+	+	?	+	+	+	+	+	
Allocation concealment (selection bias)	+	+	?	+	?	?	?	+	?	+	+	+	
Blinding of participants and personnel (performance bias)	+	+	+	+	+	+	+	+	+	+	+	+	
Blinding of outcome assessment (detection bias)	+	+	+	+	+	+	+	+	+	+	+	+	
Incomplete outcome data (attrition bias)	+	+	+	+	+	+	+	+	?	+	+	+	
Selective reporting (reporting bias)	+	+	+	+	+	+	+	+	+	+	+	+	
Other bias	+	+	+	+	+	+	+	+	+	+	+	+	

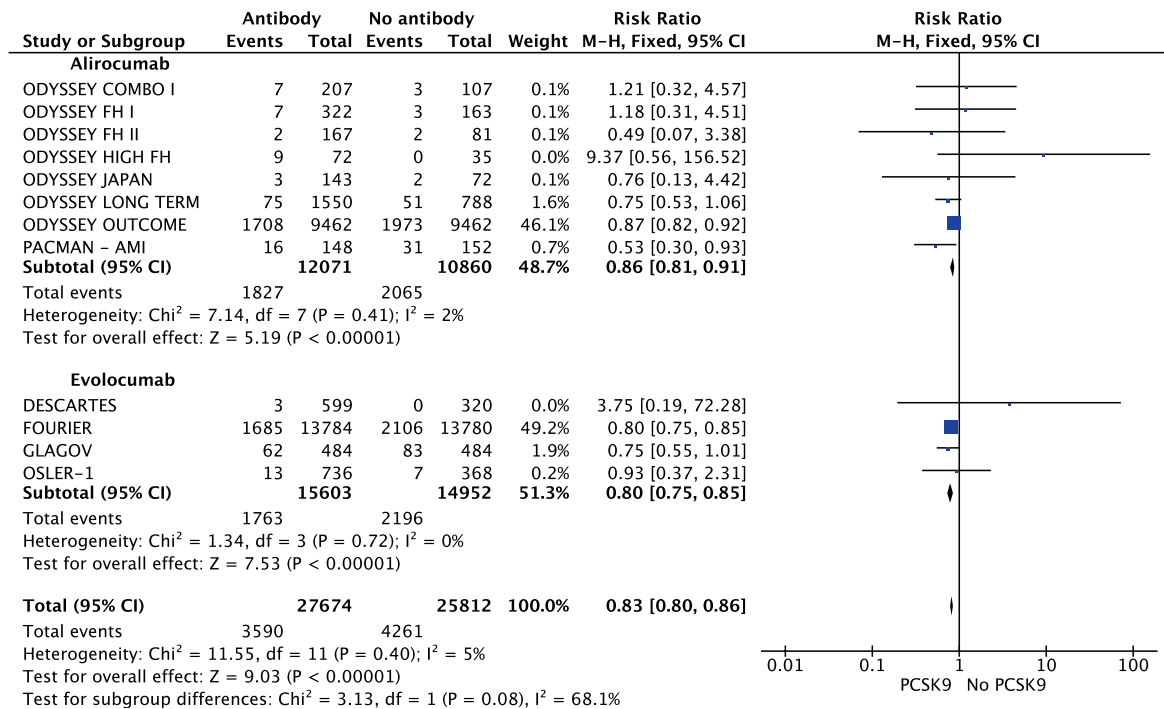
Supplementary Figure 2. Results of risk assessment.

6 Supplementary Figure 3.

(A)



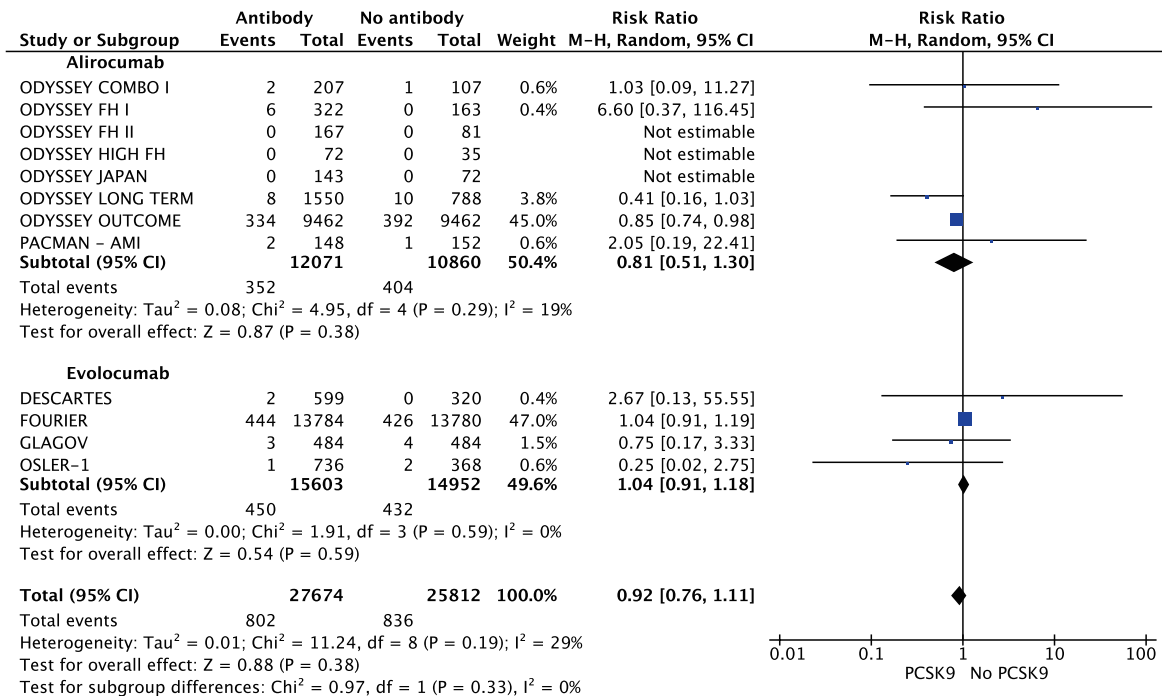
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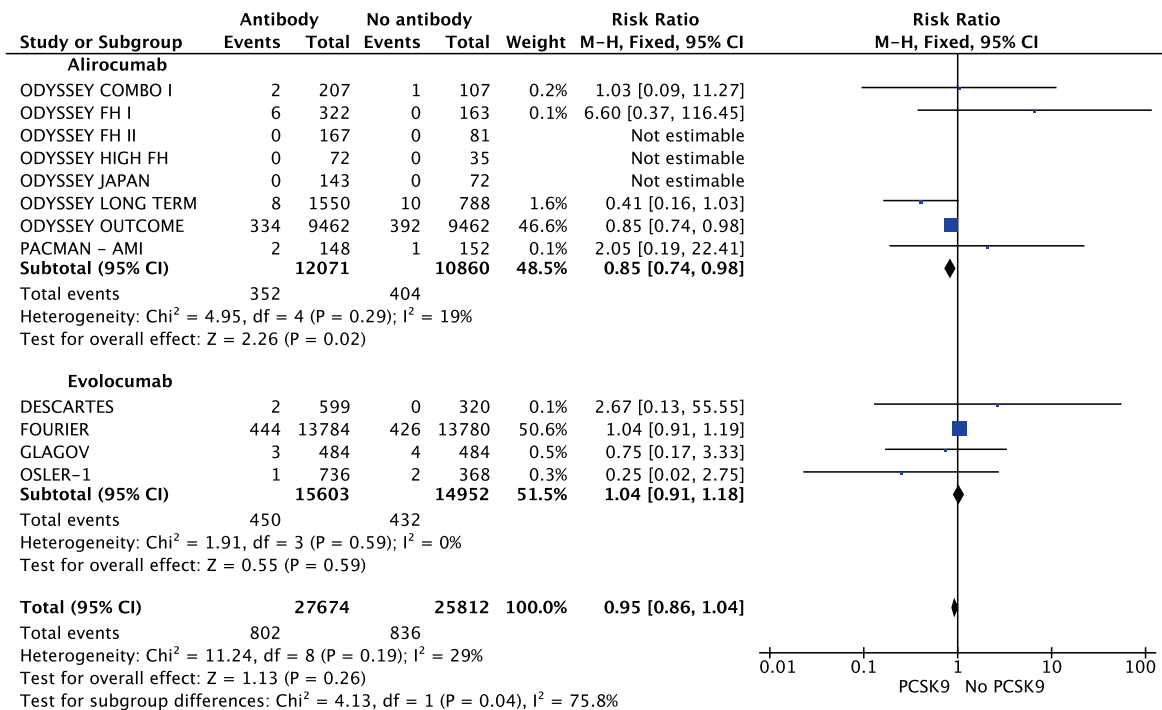
Supplementary Figure 3. Risk estimates using random effect (A) and fixed effect (B) models for major vascular events.

7 Supplementary Figure 4.

(A)



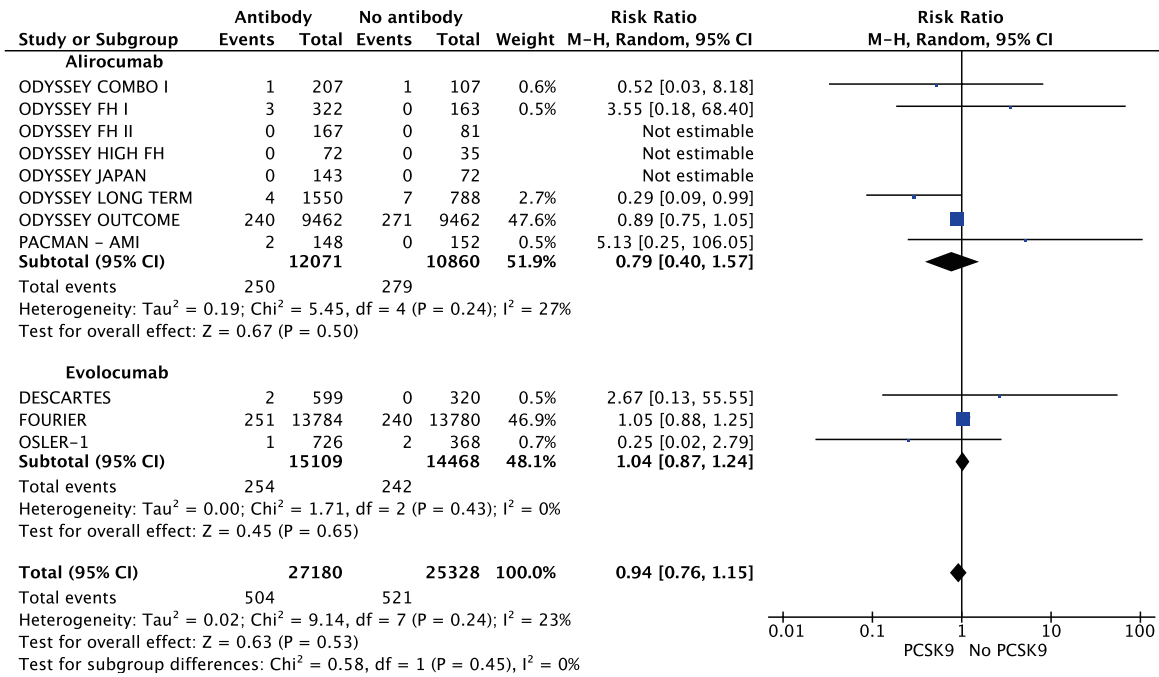
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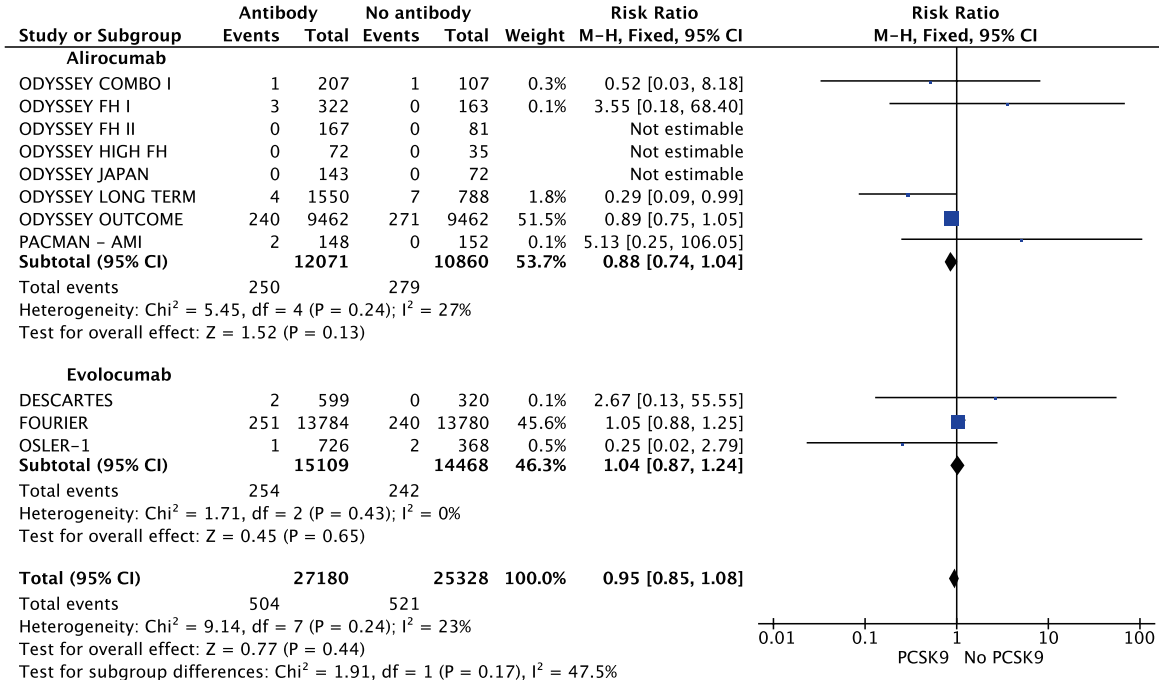
Supplementary Figure 4. Risk estimates using random effect (A) and fixed effect (B) models for all-cause death.

8 Supplementary Figure 5.

(A)



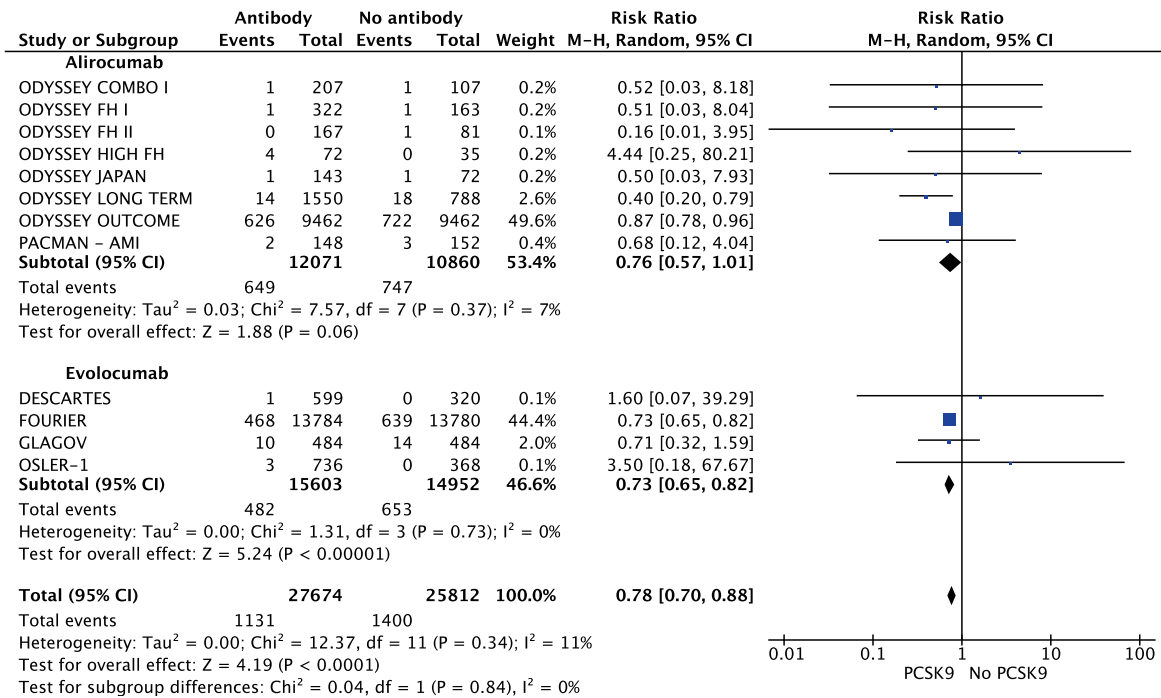
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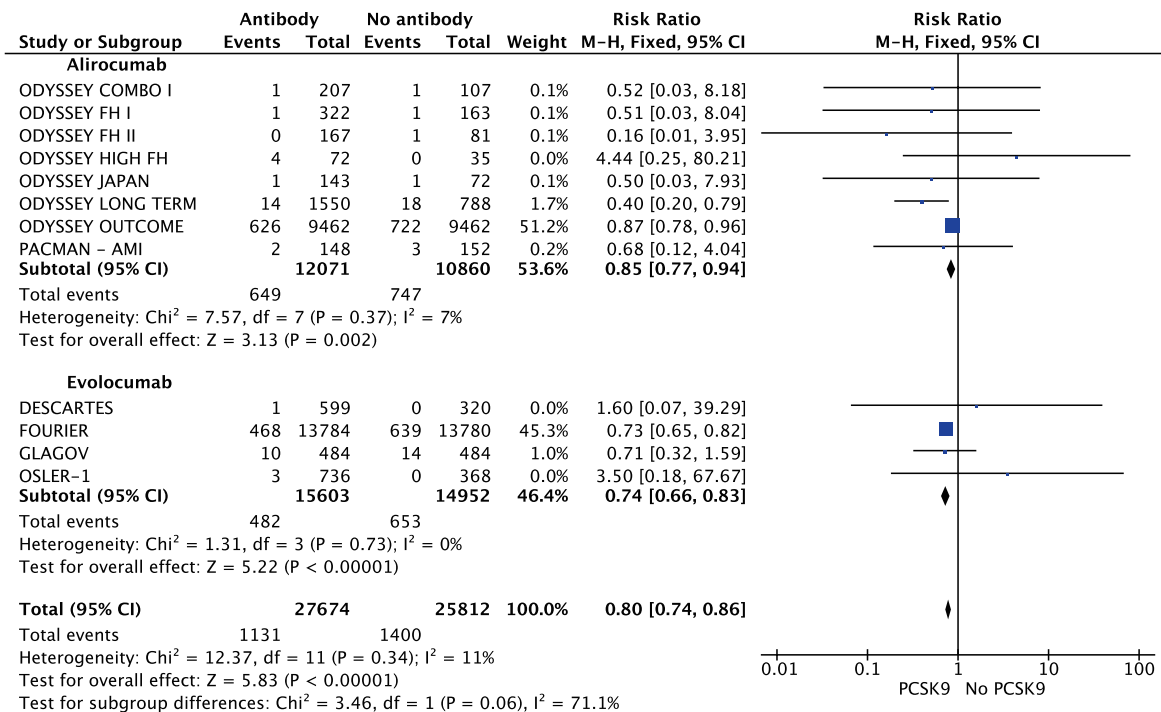
Supplementary Figure 5. Risk estimates using random effect (A) and fixed effect (B) models for cardiovascular mortality.

9 Supplementary Figure 6.

(A)



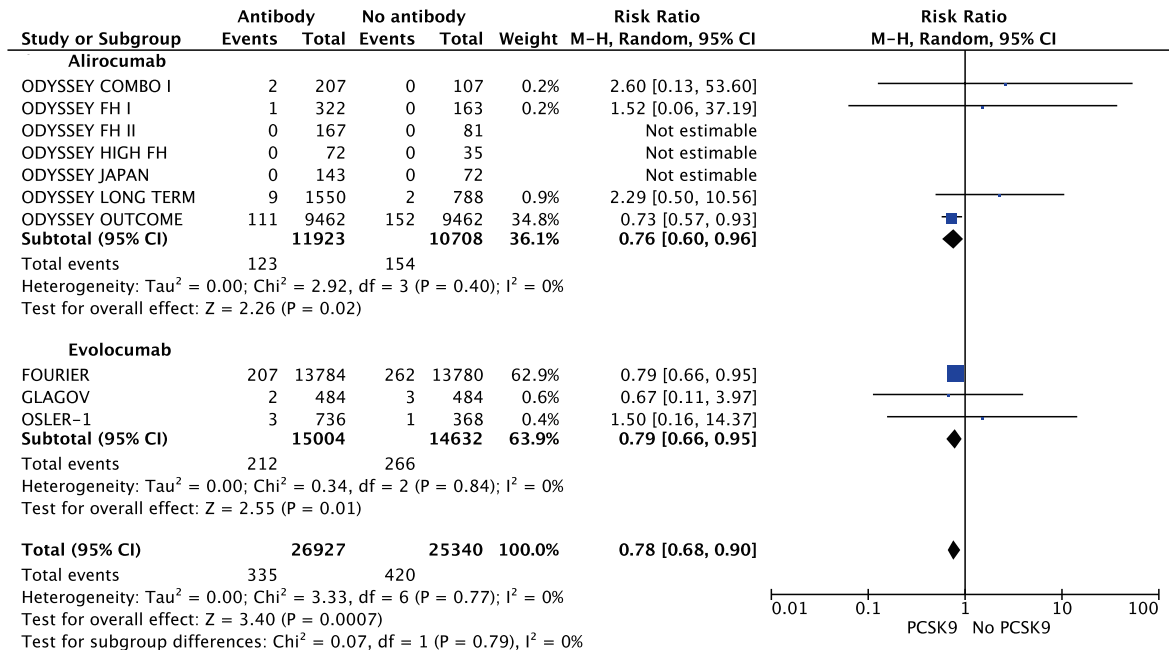
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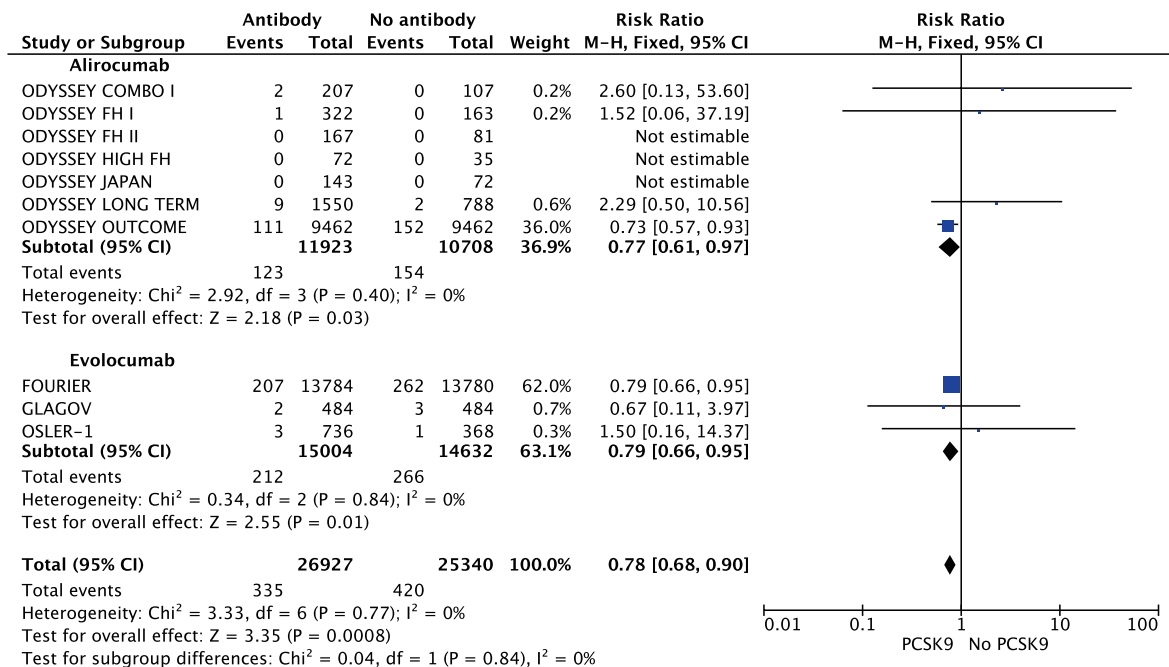
Supplementary Figure 6. Risk estimates using random effect (A) and fixed effect (B) models for myocardial infarction.

10 Supplementary Figure 7.

(A)



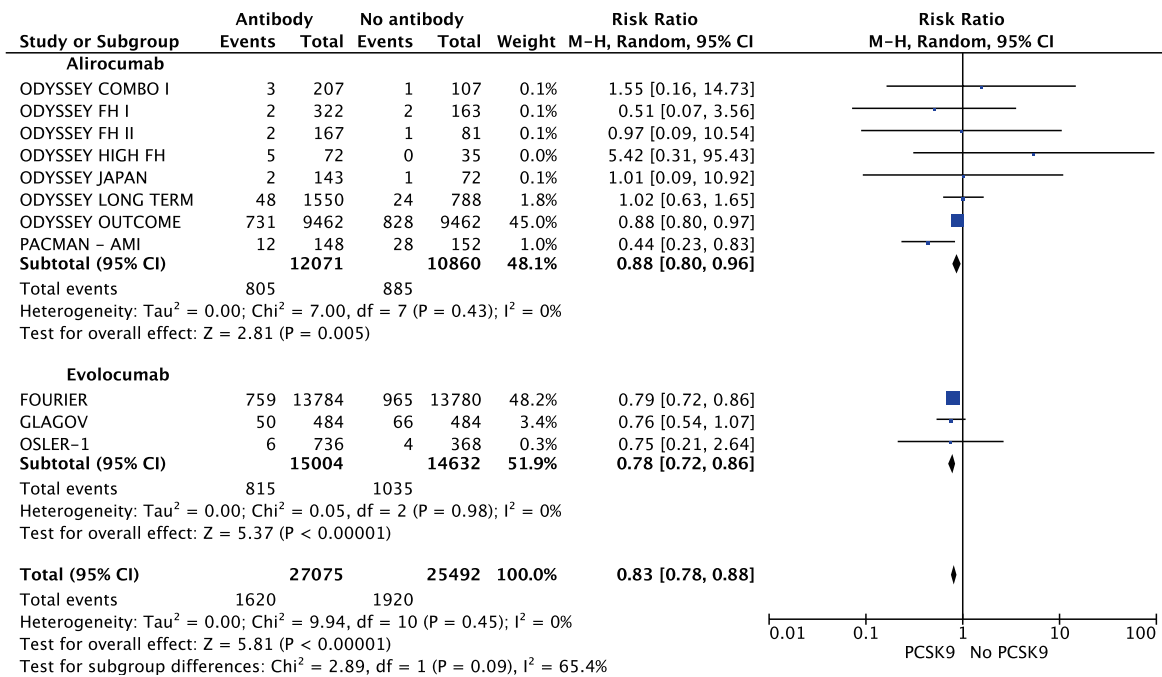
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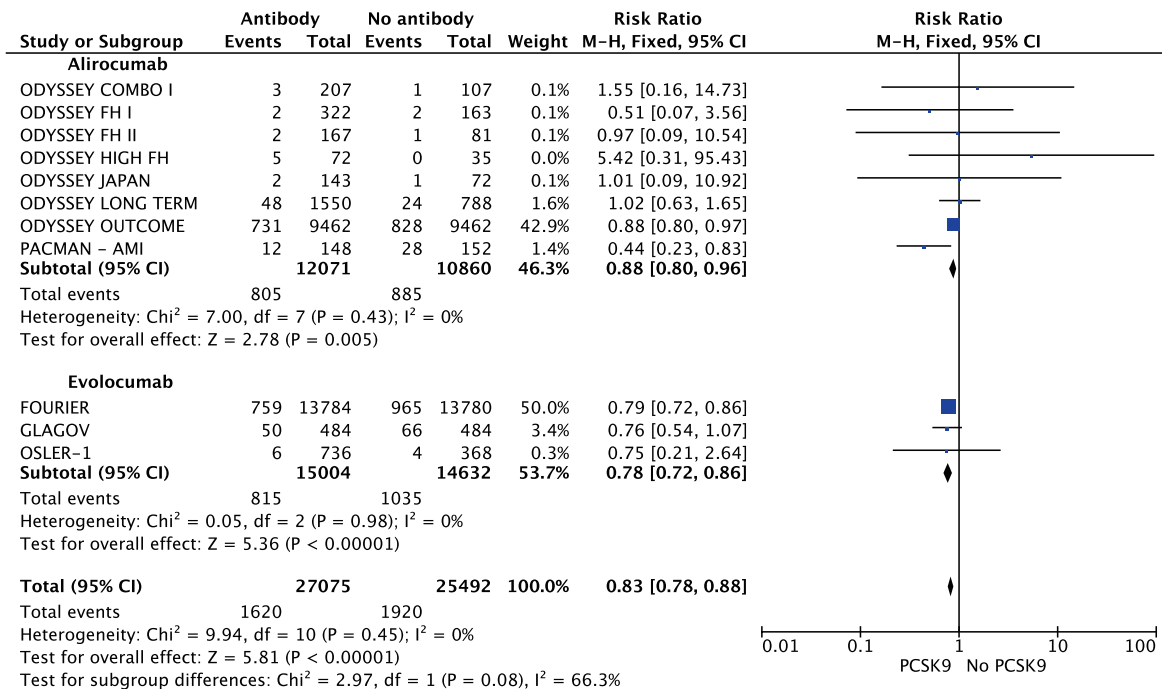
Supplementary Figure 7. Risk estimates using random effect (A) and fixed effect (B) models for stroke.

11 Supplementary Figure 8.

(A)



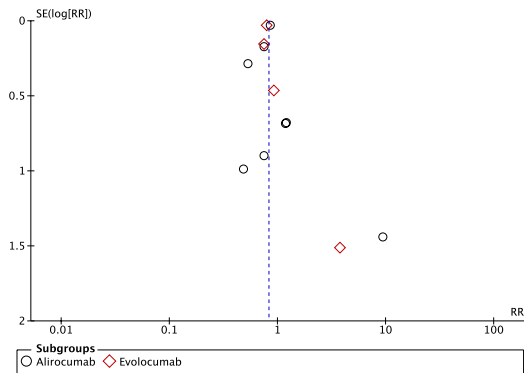
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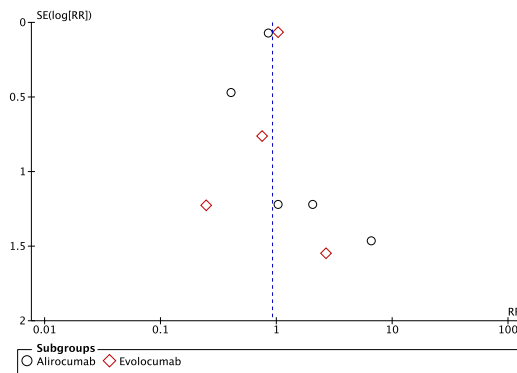
Supplementary Figure 8. Risk estimates using random effect (A) and fixed effect (B) models for coronary revascularization.

12 Supplementary Figure 9.

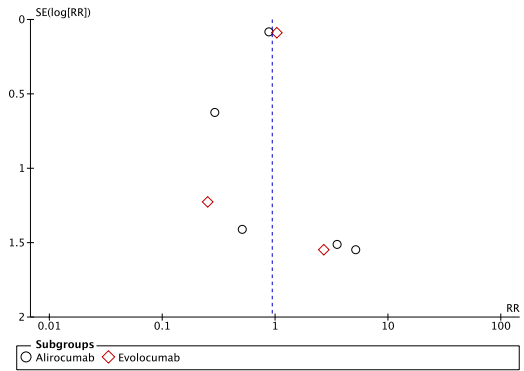
(A)



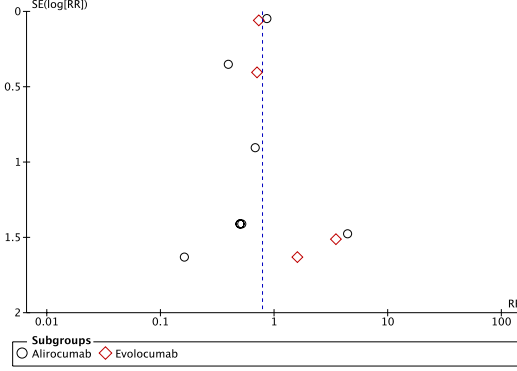
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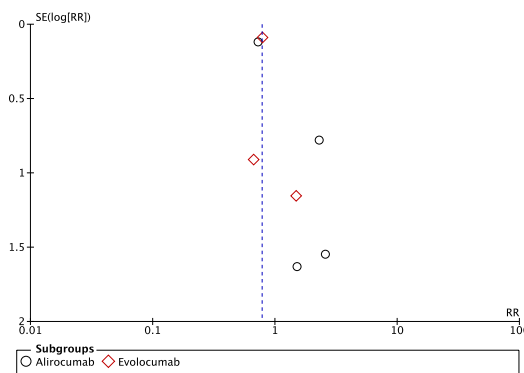
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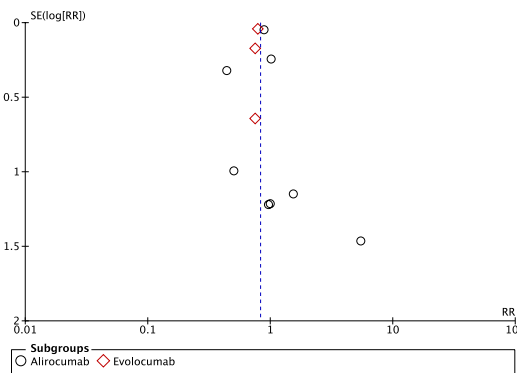
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(E)



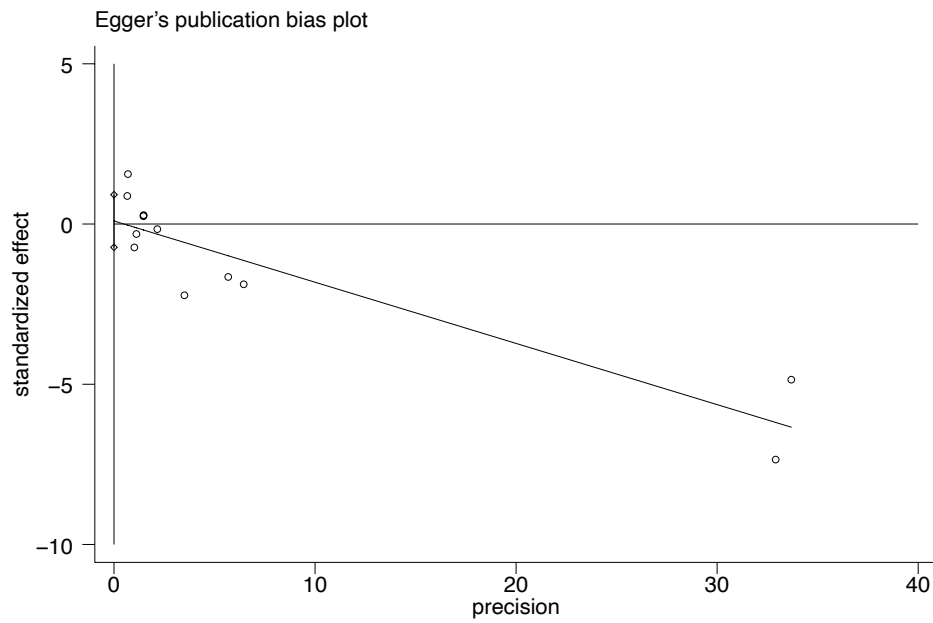
(F)



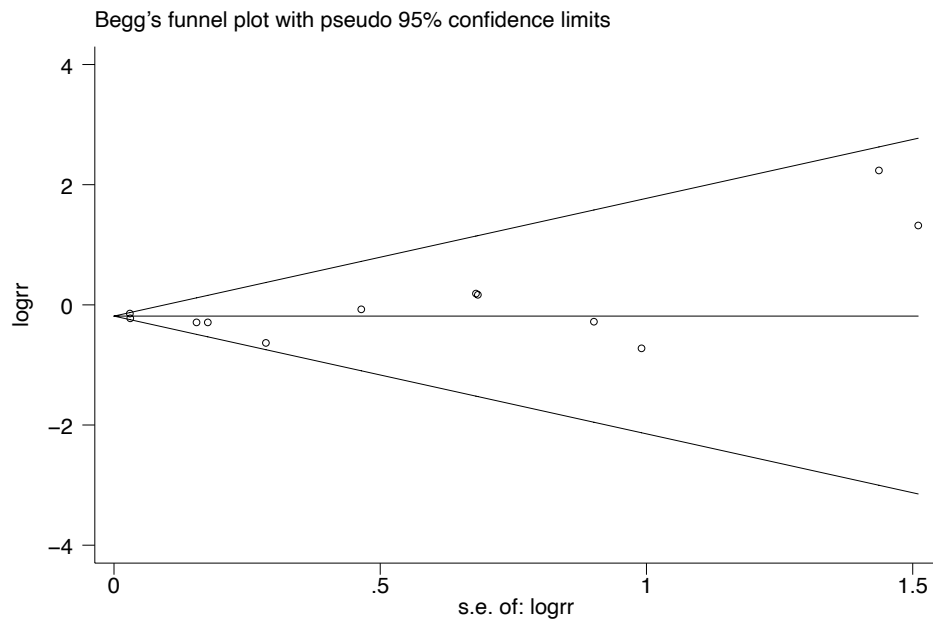
Supplementary Figure 9: Funnel plot analysis for major vascular events (A), all-cause death (B), cardiovascular mortality (C), myocardial infarction (D), stroke (E), and coronary revascularization (F).

13 Supplementary Figure 10.

(A)



(B)

**Supplementary Figure 10: Egger's test (A), Begg's test (B).**